

Amazon Elastic Beanstalk:

1. Go to **Identity and Access Management (IAM)**. Go to Roles. Create a Role. In the trusted entity type, select **AWS service**. Take **EC2** in Use Case. Go to the next page. In Permission Policies, select **AWSElasticBeanstalkWebTier**. Give a role name (say **'NewRole'**).
2. Go to **Amazon Elastic Beanstalk**. Create an application. In the Environment Tier, select the **Web server environment**. Give an application name (say **'MyApp'**). Environment name will come automatically. In the Platform, select **'Managed Platform'**. Choose **'Python'** as a Platform. We can keep the platform branch as **'Python 3.12 running on 64bit Amazon Linux 2023'**. We can keep the platform version as **'4.2.0'**. In application code, select **'Upload your code'**. Give the Version label as v1. Select **local file**. Select a .zip file having the application (say a Flask application) and upload it. In Presets, select **Single instance** as configuration presets.
3. In the Service role, select **Use an existing service role**. Select **aws-elasticbeanstalk-service-role**. In EC2 key pair, select your key pair. In the EC2 instance profile, select the role you have created in **Step 1** (Role name is **'NewRole'**).
4. In VPC, select the default VPC. In the Instance Subnets, select one availability zone (say **ap-south-1a**).
5. In Instance metadata service (IMDSv1), select **'Deactivated'**. Select one EC2 security group. In the Auto Scaling group, select **'Single instance'** as Environment type. In Fleet composition, select **On-Demand instance**. In the instance types, select **t3.micro**.
6. In Health reporting, select **'Basic'** in the System. In the managed update, do not select **'Activated'**. In the Proxy server, select **'Nginx'**. A review page will come. Go to the next page. Click the **Submit** button.
7. After a few minutes, the **Application** and the **Environment** will be created. Go to the Environments. Check the **Domain**. Open the domain in the web browser. We can find that our application is deployed in AWS using **Amazon Elastic Beanstalk**.
8. Take the necessary screenshots. After the work is done, **delete the application and terminate the environment** in **Amazon Elastic Beanstalk**. After this, EC2 instances will be terminated by Amazon Elastic Beanstalk. We can check that from the **'EC2 Dashboard'**. Also, Amazon Elastic Beanstalk will delete the security group that it created during the deployment process. We can check that from **'Security Groups'**. Go to the **Amazon S3** and delete the bucket and its contents.